MILITARY SPECIFICATION

SULFUR DIOXIDE, TECHNICAL

This specification is approved for use by the Directorate of Energy Management, Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of sulfur dioxide, technical.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MIL-STD-101 - Color Code for Pipelines and for Compressed - Gas Cylinders

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-147 - Palletized and Containerized Unit Loads

MIL-STD-1411 - Inspection and Maintenance of Compressed Gas Cylinders

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Transportation (DOT)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SA-ALC/SFRM, Kelly AFB TX 78241-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6830

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49CFR, Parts 100-199 - Transportation.

(The Code of Federal Regulations (CFR) and the Federal Register are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof.)

American Society for Testing and Materials Publications

ASTM E-29 - Recommended Practices for Designating Significant Places in Specified Limiting Values.

ASTM E-203 - Standard Test Methods for Water Using Karl Fischer Reagent.

ASTM D-1193 - Specification for Reagent Water.

(Copies of ASTM publications may be obtained upon application to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19102.)

Compressed Gas Association

V-1 American-Canadian Standard Compressed Gas Cylinder Valve Outlet and Inlet Connection; ANSI 57.1; CSA - B96.

(Copies of CGA publications may be obtained from Compressed Gas Association, Inc., 1235 Jefferson Davis Hwy, Arlington VA 22202.

American Chemical Society (ACS) Specification

ACS Reagent Chemical Standards.

(Application for copies should be addressed to the American Chemical Society, 1155 16th St. NW, Washington DC 20036.)

2.2.1 Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 Material. The sulfur dioxide shall conform to the requirements specified in Table I, when tested in accordance with the applicable test method.

3.2 Limiting Values. The following applies to all specified limits of this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limitation value, in accordance with the rounding-off method of the Recommended Practices and for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

3.3 Cylinder Maintenance. Government-furnished cylinders that require maintenance shall be processed by the supplier for serviceability to meet the requirements of this specification and 49CFR171-190 and MIL-STD-1411 (6.2).

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TABLE I

Physical and Chemical Requirements		
Characteristic	Properties	Test Paragraph
Purity by difference, % by wt	99.98	4.6.4
Water, % by wt	0.010	4.6.3
Residue, % by wt	0.005	4.6.1
Acid as % H_2SO_4 by wt	0.002	4.6.2

3.4 Valves. Replacement valve for defective valves in Government-furnished cylinder shall be supplier-furnished and shall be in accordance with Compressed Gas Association Connection No. 668.

3.5 Leakage. Cylinders and valves shall not leak after being filled when tested in accordance with 4.7.1.

3.6 Rejected Cylinders. Government-owned cylinders considered unsatisfactory for filling shall be processed as specified in the contract or purchase order (6.2). Other cylinders shall be disposed of by the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (6.2).

4.1.1 Points of Inspection (6.2)

4.1.1.1 Cylinders. Unless otherwise specified, acceptance tests of sulfur dioxide in cylinders shall be conducted at the sites of filling. The acceptance tests shall be completed prior to shipment of cylinders from the shipping site.

4.1.1.2 Containers. When bulk containers are filled from manufacturer's plant system or other storage facilities for delivery to destination by any mode of transportation, acceptance test shall be conducted on the container contents prior to leaving the site of filling.

4.2 Classification of Tests. The inspection and testing of the product shall be classified as quality conformance tests.

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ection 5. Defects shall be scored in accordance with [illegible] final packaging, packing and marking comply with s

4.3 Quality Conformance Tests. Quality conformance tests shall consist of:

a. Individual Tests 4.3.1

b. Sampling Tests 4.3.2

4.3.1 Individual Tests. Each cylinder shall be subjected to the following test:

Leakage test 4.7.1

4.3.2 Sampling Tests. The product shall be selected in accordance with 4.3.2.1 and subjected to the following tests as described under 4.6.

4.3.2.1 Sampling Plan. Quality of the product shall normally be verified by analysis as samples by lot as specified herein.

4.3.2.1.1 Lotting. A lot shall consist of one of the following:

a. The product produced in not more than 24 consecutive hours from a continuous process which is used to fill shipping cylinders directly from the process output. A continuous process shall be the production of product by continuous input of raw materials and output of finished product by one manufacturer in one plant with no change in manufacturing conditions or materials.

b. The product from individual runs of a batch process which is used to fill shipping cylinders directly from the process output. A batch process shall be the production of product by runs from single additions of raw materials which are mixed, reacted, and purified as necessary forming the product.

c. The product from either or both the continuous and batch processes which is held in a single storage tank and subsequently withdrawn to fill shipping cylinders. The product shall be mixed until homogeneous at the time of withdrawal and shall not be added to while being withdrawn. After each addition to the storage tank, the contents shall constitute a separate lot.

4.3.2.1.2 Sample. Each sample shall be of sufficient size to conduct all the quality conformance tests as specified herein. Unless otherwise specified, the quality conformance tests shall be made of each required sample (6.2). When required, an equivalent sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the quality conformance tests specified herein.

4.3.2.1.3 Cylinders. The number of cylinders selected for sampling from each lot shall be in accordance with Table II. Also, the first and last cylinders to be filled within a given lot shall be sampled. Other samples shall be selected at random. The product from each cylinder sampled shall constitute a separate sample.

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TABLE II

Sampling Plan for Test	
Number of Cylinders in Lot	Number of Cylinders to be Sampled
1 - 9	1
10 - 49	2
50 - 149	3
150 and over	4

4.3.2.1.4 Bulk Transports. Sample shall be taken from each portable tank, cargo tank, or tank car filled with sulfur dioxide.

4.3.2.1.5 Other Containers (6.2). Unless otherwise specified, other containers of 100 gallons or less water capacity shall be sampled in accordance with 4.3.2.1.3. Containers greater than 100 gallons water capacity shall be sampled in accorance with 4.3.2.1.4.

4.4 Rejection. When any samples of the product tested as specified herein fail to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected.

4.5 Chemicals. Unless otherwise specified, all chemical tests shall be made with ACS specification reagent grade chemicals and ASTM D-1193, Type III, Reagent Water.

4.6 Test Method.

4.6.1 Residue. The residue content of the sulfur dioxide shall be determined by one of the following methods. (4.6.1.1 is the referee method)

4.6.1.1 Transfer 100 ml of the sample into a clean 125 ml erlenmeyer flask weighed to nearest .0001 gram. Evaporate the sample to dryness on a steam bath. After the liquid has evaported, displace the sulfur dioxide vapors with dry air, wipe the flask dry and place in a desiccator to cool for 30 minutes and reweigh.

$$\% \text{ Residue} = \frac{\text{Diff in wt (grams)} \times 100}{100 \text{ ml} \times 1.44 \text{ g/ml}}$$

4.6.1.2 Residue content may be determined by any method regularly used by the supplier and approved by the procuring activity.

4.6.2 Acid as Sulfuric Acid. The acid content of the sulfur dioxide shall be determined by one of the following methods. (4.6.2.1 is the referee method)

4.6.2.1 Add 25 ml distilled water to residue 4.6.1 and titrate with .01N NaOH using methyl orange as an indicator.

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$$\% H_2SO_4 = \frac{ml\ NaOH \times .01N \times .049 \times 100}{100 \times 1.44}$$

4.6.2.2 Sulfuric Acid content may be determined by any method regularly used by the supplier and approved by the procuring activity.

4.6.3 Moisture. The moisture content of the sulfur dioxide shall be determined by one of the following methods. (4.6.3.1 is the referee method)

4.6.3.1 ASTM Method E203-62 may be used.

4.6.3.2 Moisture content may be determined by any method regularly used by the supplier and approved by the procuring activity.

4.6.4 Purity.

Purity = 100 - (% Residue + % Water)

4.7 Cylinder Test.

4.7.1 Leakage. Each cylinder shall be leak tested IAW MIL-STD-1411.

4.8 Inspection of Preparation for Delivery. The packing, packaging, and marking for shipment and storage of the material shall be inspected to determine compliance with the requirements of Section 5 of the specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A, B, and C. Unless otherwise specified, the sulfur dioxide shall be packaged in government-furnished cylinders in accordance with 49 CFR, 171-190. Cylinders shall conform to DOT Specification 3A225, 3AA225, 3B225, 4A225, 4B225, or 4BA225. One-ton containers shall conform to DOT 106-A-500.

5.2 Packing.

5.2.1 Level A, B, and C. When specified (6.2) cylinders shall be palletized in accordance with MIL-STD-147.

5.3 Marking. Cylinders or containers shall be marked in accordance with 49 CFR, 171-190 and MIL-STD-129. Color coding shall be in accordance with MIL-STD-101 (6.2).

5.3.1 Identification Tag. Unless otherwise specified in the contract or purchase order, an identification tag impervious to climatic conditions shall be wired to the outlet port of each cylinder and shall contain the following information: product name, specification number with revision letter, federal stock number, quantity, name of manufacturer, name of contractor (if different from manufacturer), date of manufacture and lot identification number.

6. NOTES

6.1 Intended Use. The sulfur dioxide covered by this specification is intended for use in treating industrial wastes.

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6.2 Ordering Data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number, and date of this specification.

b. Method of shipment, type and capacity of cylinders (5.1).

c. When inspection requirements are to be performed by other than the supplier (4.1).

d. When variation in points of inspection are granted (4.1.1).

e. When cleaning and repair schedule is required for government-owned cylinders (3.3).

f. When instructions for disposition of rejected cylinders are required (3.6).

g. When identification tag is to be other than as specified (5.3).

h. When supplier-furnished cylinders are required (5.1).

i. When the cylinder color code is to be waived. (5.2.1)

6.3 Subject Term (key word) Listing: sulfur dioxide.

Custodians:

Air Force - 68

Review Activity:

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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